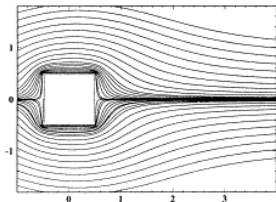


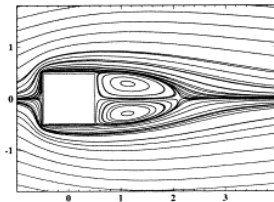
Equations to be solved

$$\nabla \cdot \mathbf{u} = 0$$

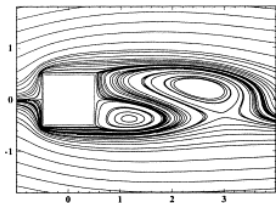
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}$$



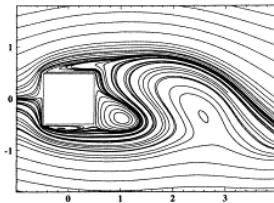
(a)



(b)



(c)



(d)

LBM algorithm

- Initialize DFs e.g. $f_i^{\text{eq}}(\rho_{\text{ini}}, \mathbf{u}_{\text{ini}})$
- Setup τ according to material properties
- Repeat until steady state or desired time is achieved
 1. Compute macroscopic quantities from f_i
 2. Compute equilibrium DF's and post-collision state
 3. Stream f_i^* along lattice links
 4. Apply BC

$$f_i(t+1, \mathbf{x} + \mathbf{c}_i) = f_i(t, \mathbf{x}) + \mathcal{M}^{-1} \mathcal{S}(m_i^{\text{eq}}(t, \mathbf{x}) - m_i(t, \mathbf{x})) = f_i^*$$

$$m_i^{\text{eq}} = \mathcal{M} f_i^{\text{eq}} \quad m_i = \mathcal{M} f_i$$

$$\nu_{lb} = c_s^2 \left(\tau - \frac{1}{2} \right),$$

$$f_i^{\text{eq}}(t, \mathbf{x}) = w_i \rho(t, \mathbf{x}) \left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}(t, \mathbf{x})}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right)$$

macro-variables...

$$\rho = \sum_i f_i(t, \mathbf{x}) \quad \rho \mathbf{u} = \sum_i \mathbf{c}_i f_i(t, \mathbf{x})$$

Multiple Relaxation Time LBM

Stability of SRT can be improved by performing collision in moment space,

$$f_i(x + c_i \delta t, t + \delta t) - f_i(x, t) = -\mathbb{M}^{-1} \mathbb{S} [\mathbf{m}(x, t) - \mathbf{m}^{\text{eq}}(x, t)]$$

$\mathbf{m} = (m_0, m_1, \dots, m_n)^T$ are vectors of moments, $\mathbb{S}$ is diagonal matrix of relaxation times for each moment and mapping between distribution and moment spaces is given by

$$\mathbf{m} = \mathbb{M} \mathbf{f}, \quad \mathbf{f} = \mathbb{M}^{-1} \mathbf{m}$$

for D2Q9 model the vector of moments reads

$$\mathbf{m} = (\rho, e, \epsilon, j_x, q_x, j_y, q_y, p_{xx}, p_{xy})^T$$

and the equilibrium moment vector reads

$$\mathbf{m}^{\text{eq}} = (\rho, -2\rho + 3(j_x^2 + j_y^2), \rho - 3(j_x^2 + j_y^2), j_x, -j_x, j_y, -j_y, (j_x^2 - j_y^2), j_x j_y)^T$$

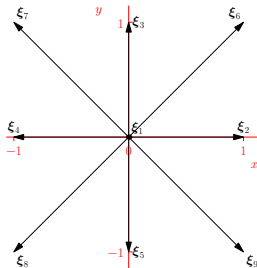
MRT Matrix

	C	E	N	W	S	NE	NW	SW	SE
m_ρ	1	1	1	1	1	1	1	1	1
m_e	-4	-1	-1	-1	-1	2	2	2	2
m_ϵ	4	-2	-2	-2	-2	1	1	1	1
m_{j_x}	0	1	0	-1	0	1	-1	-1	1
m_{q_x}	0	-2	0	2	0	1	-1	-1	1
m_{j_y}	0	0	1	0	-1	1	1	-1	-1
m_{q_y}	0	0	-2	0	2	1	1	-1	-1
$m_{p_{xx}}$	0	1	-1	1	-1	0	0	0	0
$m_{p_{xy}}$	0	0	0	0	0	1	-1	1	-1

MRT Inverse Matrix

	m_ρ	m_e	m_ϵ	m_{j_x}	m_{q_x}	m_{j_y}	m_{q_y}	$m_{p_{xx}}$	$m_{p_{xy}}$
f_C	1/9	-1/9	1/9	0	0	0	0	0	0
f_E	1/9	-1/36	-1/18	1/6	-1/6	0	0	1/4	0
f_N	1/9	-1/36	-1/18	0	0	1/6	-1/6	-1/4	0
f_W	1/9	-1/36	-1/18	-1/6	1/6	0	0	1/4	0
f_S	1/9	-1/36	-1/18	0	0	-1/6	1/6	-1/4	0
f_{NE}	1/9	1/18	1/36	1/6	1/12	1/6	1/12	0	1/4
f_{NW}	1/9	1/18	1/36	-1/6	-1/12	1/6	1/12	0	-1/4
f_{SW}	1/9	1/18	1/36	-1/6	-1/12	-1/6	-1/12	0	1/4
f_{SE}	1/9	1/18	1/36	1/6	1/12	-1/6	-1/12	0	-1/4

LBM D2Q9 Model



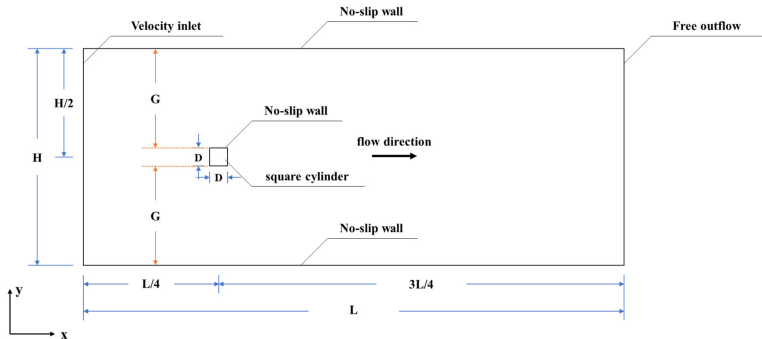
	C	E	N	W	S	NE	NW	SW	SE
c_i	(0,0)	(1,0)	(0,1)	(-1,0)	(0,-1)	(1,1)	(-1,1)	(-1,-1)	(1,-1)
t_i	4/9	1/9	1/9	1/9	1/9	1/36	1/36	1/36	1/36

lattice speed of sound $c_s = \frac{1}{\sqrt{3}}$

Physical & lattice units relations

Reynolds and Mach numbers are used to setup the LBM parameters

$$u_{x,lb} = Ma\sqrt{c_s^2} \quad \nu_{lb} = \frac{u_{x,lb}L_{Y,lb}}{Re}$$



Initial and boundary conditions

Initial conditions:

$$\rho_{lb}(t = 0, \mathbf{x}) = 1 \quad \mathbf{u}_{lb}(t = 0, \mathbf{x}) = (0, 0)$$

Boundary conditions:

$$\rho_{lb,in} = 1 \quad \mathbf{u}_{lb,in} = Ma \cdot c_s$$

$$\mathbf{u}_{wall} = 0 \quad (\text{BounceBack halfway})$$

$$\rho_{out} = \rho(\mathbf{x}_{out} - dx\mathbf{n}_{out}) \quad \mathbf{u}_{out} = \mathbf{u}(\mathbf{x}_{out} - dx\mathbf{n}_{out})$$